WORKED SOLUTION

STEP II 2020, Question 6

Pure • Matrix Powers and Trace

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Read the question before you go on. It is in the STEP II paper in OCR's download of the 2020 STEP papers (a zip file):

www.ocr.org.uk/Images/696589-2020-past-papers.zip

This question has a full guided treatment in *Beyond STEP*, Section 7.4, *Matrices That Return to the Start*: https://beyondstep.keithmcnulty.org/ch07/finite-order.html

The key idea

Every real 2×2 matrix satisfies its own characteristic equation,

$$\mathbf{M}^2 - t\,\mathbf{M} + \delta\,\mathbf{I} = \mathbf{0}, \qquad t = \operatorname{tr}(\mathbf{M}),\ \delta = \det(\mathbf{M}),$$

the Cayley–Hamilton theorem (it is a two-line check by multiplying out). Once $\mathbf{M}^2$ is written in terms of $\mathbf{M}$ and $\mathbf{I}$, any equation such as $\mathbf{M}^2 = \mathbf{I}$ becomes "$t\mathbf{M}$ is a multiple of $\mathbf{I}$", which forces either $t = 0$ or $\mathbf{M}$ itself to be a multiple of $\mathbf{I}$. Parts (iii) and (iv) then come from applying part (ii) to $\mathbf{M}^2$ and $\mathbf{M}^4$ instead of $\mathbf{M}$.

Throughout, write $\mathbf{M} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, $t = a + d$ and $\delta = ad - bc$.

Solution

Part (i)

Multiplying out,

$$\mathbf{M}^2 = \begin{pmatrix} a^2 + bc & ab + bd \\ ca + dc & bc + d^2 \end{pmatrix} = (a+d)\begin{pmatrix} a & b \\ c & d \end{pmatrix} - (ad - bc)\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix},$$

since, for example, $(a+d)a - (ad-bc) = a^2 + bc$ and $(a+d)b = ab + bd$. So

$$\mathbf{M}^2 = t\mathbf{M} - \delta\mathbf{I}. \qquad (*)$$

Taking the trace of $(*)$, and using $\operatorname{tr}(\mathbf{I}) = 2$,

$$\operatorname{tr}(\mathbf{M}^2) = t \cdot t - 2\delta = \operatorname{tr}(\mathbf{M})^2 - 2\det(\mathbf{M}),$$

as required. (Directly: $a^2 + 2bc + d^2 = (a+d)^2 - 2(ad - bc)$.)

Part (ii)

First equivalence, (⇒). Suppose $\mathbf{M}^2 = \mathbf{I}$ and $\mathbf{M} \neq \pm\mathbf{I}$. Substituting into $(*)$ gives

$$t\,\mathbf{M} = (1+\delta)\,\mathbf{I}.$$

If $t \neq 0$ we could divide to get $\mathbf{M} = k\mathbf{I}$ with $k = (1+\delta)/t$ real; then $\mathbf{M}^2 = k^2\mathbf{I} = \mathbf{I}$ gives $k = \pm 1$, so $\mathbf{M} = \pm\mathbf{I}$, which is excluded. Hence $t = 0$, and then $(1+\delta)\mathbf{I} = \mathbf{0}$, so $\delta = -1$.

(⇐). If $t = 0$ and $\delta = -1$, then $(*)$ gives $\mathbf{M}^2 = 0 \cdot \mathbf{M} + \mathbf{I} = \mathbf{I}$. Also $\operatorname{tr}(\pm\mathbf{I}) = \pm 2 \neq 0$, so $\mathbf{M} \neq \pm\mathbf{I}$.

Second equivalence, (⇒). If $\mathbf{M}^2 = -\mathbf{I}$, then $(*)$ gives $t\mathbf{M} = (\delta - 1)\mathbf{I}$. If $t \neq 0$ then $\mathbf{M} = k\mathbf{I}$ for a real k, and $\mathbf{M}^2 = k^2\mathbf{I} = -\mathbf{I}$ would need $k^2 = -1$, impossible for real k. So $t = 0$, and then $\delta = 1$.

(⇐). If $t = 0$ and $\delta = 1$, then $(*)$ gives $\mathbf{M}^2 = -\mathbf{I}$ at once.

Ask yourself Where exactly did we use that $\mathbf{M}$ is *real*?

Only in the second equivalence: $k^2 = -1$ has no real solution. For complex matrices, $\mathbf{M} = \mathrm{i}\mathbf{I}$ satisfies $\mathbf{M}^2 = -\mathbf{I}$ with trace $2\mathrm{i} \neq 0$, so the result would be false.

Part (iii)

(⇐) is immediate: if $\mathbf{M}^2 = \pm\mathbf{I}$ then $\mathbf{M}^4 = (\pm\mathbf{I})^2 = \mathbf{I}$.

(⇒). Let $\mathbf{N} = \mathbf{M}^2$, a real 2×2 matrix with $\mathbf{N}^2 = \mathbf{I}$. Either $\mathbf{N} = \pm\mathbf{I}$, which is what we want, or $\mathbf{N} \neq \pm\mathbf{I}$, in which case part (ii) says $\det(\mathbf{N}) = -1$. But

$$\det(\mathbf{N}) = \det(\mathbf{M})^2 \geq 0,$$

so the second case cannot happen. Hence $\mathbf{M}^2 = \pm\mathbf{I}$.

The condition for $\mathbf{M}^4 = -\mathbf{I}$. Again put $\mathbf{N} = \mathbf{M}^2$. By the second equivalence of part (ii) applied to $\mathbf{N}$,

$$\mathbf{M}^4 = -\mathbf{I} \iff \operatorname{tr}(\mathbf{N}) = 0 \text{ and } \det(\mathbf{N}) = 1.$$

Using part (i) and $\det(\mathbf{N}) = \delta^2$, this says

$$t^2 - 2\delta = 0 \quad \text{and} \quad \delta^2 = 1.$$

Now $\delta^2 = 1$ gives $\delta = \pm 1$, and $t^2 = 2\delta \geq 0$ rules out $\delta = -1$. So $\delta = 1$ and $t^2 = 2$. Every step was an equivalence, so

$$\boxed{\mathbf{M}^4 = -\mathbf{I} \iff \det(\mathbf{M}) = 1 \text{ and } \operatorname{tr}(\mathbf{M}) = \pm\sqrt{2}}.$$

Part (iv)

Applying part (iii) to $\mathbf{M}^2$: $\mathbf{M}^8 = \mathbf{I} \iff \mathbf{M}^4 = \pm\mathbf{I}$, so the matrices with $\mathbf{M}^8 = \mathbf{I}$ are exactly those with

$$\mathbf{M}^2 = \pm\mathbf{I} \qquad \text{or} \qquad \det(\mathbf{M}) = 1,\ \operatorname{tr}(\mathbf{M}) = \pm\sqrt{2}.$$

Rotations and reflections are the *orthogonal* matrices: their columns are perpendicular unit vectors (rotations have determinant 1, reflections determinant -1). So we just need a matrix in this list whose columns are not unit vectors. For example,

$$\mathbf{M} = \begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix}$$

has trace 0 and determinant -1, so $\mathbf{M}^2 = \mathbf{I}$ by part (ii) (check: $\mathbf{M}^2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$), and hence $\mathbf{M}^8 = \mathbf{I}$. Its second column $(1, -1)$ has length $\sqrt{2} \neq 1$, so it is neither a rotation nor a reflection. (It is an "oblique reflection": it fixes every vector along the x-axis and reverses every vector along $(1, -2)$, and these two directions are not perpendicular.)

An example from the more interesting family is $\mathbf{M} = \begin{pmatrix} \sqrt{2} & 1 \\ -1 & 0 \end{pmatrix}$: determinant 1 and trace $\sqrt{2}$, so $\mathbf{M}^4 = -\mathbf{I}$ and $\mathbf{M}^8 = \mathbf{I}$, yet its first column has length $\sqrt{3}$. It is a "distorted" rotation through 45°: it has order exactly 8, like the rotation, but does not preserve lengths.

Checks

- The rotation through 45°, $\frac{1}{\sqrt{2}}\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$, has determinant 1 and trace $\sqrt{2}$, and four 45° turns make a half-turn, which is $-\mathbf{I}$: consistent with part (iii).
- Squaring $\mathbf{M} = \begin{pmatrix} \sqrt{2} & 1 \\ -1 & 0 \end{pmatrix}$ gives $\begin{pmatrix} 1 & \sqrt{2} \\ -\sqrt{2} & -1 \end{pmatrix}$, which has trace 0 and determinant 1, so its square is $-\mathbf{I}$ by part (ii).
- A computer search over all matrices with entries in $\{-3, -\frac{5}{2}, \ldots, 3\}$ confirmed both equivalences in part (ii) and the first in part (iii) exactly.

Common slips

- Dividing by t without first handling $t = 0$; the whole argument in part (ii) is a case split on whether t is zero.
- In part (iii), forgetting that $\det(\mathbf{M}^2) = \det(\mathbf{M})^2$ cannot be negative; that is precisely what rules out the "other" square roots of $\mathbf{I}$.
- Giving a rotation through 45° as the example in part (iv): it satisfies $\mathbf{M}^8 = \mathbf{I}$ but is exactly what is excluded.

Beyond STEP

The identity $(*)$ is the 2×2 case of the Cayley–Hamilton theorem: every square matrix satisfies its characteristic polynomial. Questions like "which $\mathbf{M}$ satisfy $\mathbf{M}^n = \mathbf{I}$?" are

really about eigenvalues: here they must be eighth roots of unity, and a real matrix of order 8 is *similar* to a rotation without being one. Strang's *Introduction to Linear Algebra* develops eigenvalues, similarity and the characteristic polynomial in this spirit.

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